

DALLAS FORT WORTH, TX, USA

WMO: 722590

Lat: 32.898N

Lon: 97.019W

Elev: 171

StdP: 99.29

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03927

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.8	-2.5	-12.8	1.3	-0.7	-10.6	1.6	2.2	13.2	9.8	12.2	11.2	5.1	340	0.538

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.6	38.6	23.4	37.3	23.6	36.1	23.7	25.9	33.1	25.5	32.7	25.0	32.2	4.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.4	28.8	23.5	18.7	28.4	23.0	18.2	28.1	81.0	33.2	78.7	32.7	76.8	32.2	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.8	10.7	9.3	-8.1	40.3	1.9	1.8	-9.4	41.6	-10.5	42.6	-11.6	43.6	-13.0	44.9		
			-9.2	26.8	1.7	0.8	-10.4	27.4	-11.4	27.8	-12.4	28.3	-13.6	28.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.6	8.4	10.6	14.8	19.0	23.7	28.0	30.1	30.2	26.3	20.3	14.2	9.2
	DBStd	8.92	5.61	5.95	5.32	4.10	3.57	2.44	2.19	2.52	3.52	4.51	5.16	5.30
	HDD10.0	277	97	59	19	2	0	0	0	0	0	1	19	79
	HDD18.3	1174	311	223	133	40	6	0	0	0	1	32	142	286
	CDD10.0	3786	47	75	167	272	424	541	623	627	491	321	144	54
	CDD18.3	1642	2	6	22	61	171	291	365	369	242	93	17	3
	CDH23.3	19503	15	48	159	460	1602	3507	5001	5072	2694	818	108	19
	CDH26.7	9839	1	8	27	101	603	1752	2839	2927	1287	276	18	2
Wind	WSAvg	4.7	4.9	5.0	5.4	5.6	5.2	4.8	4.3	3.9	3.8	4.4	4.7	4.6
Precipitation	PrecAvg	878	53	58	74	83	119	83	57	56	76	96	62	60
	PrecMax	1589	157	188	188	273	431	282	283	174	322	398	250	222
	PrecMin	520	0	4	2	3	9	9	0	0	2	0	1	4
	PrecStd	225	39	41	47	50	79	64	49	43	64	80	50	46
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.4	27.6	29.5	31.4	35.2	37.8	39.9	40.6	38.5	33.6	28.9	25.9
		MCWB	15.7	16.1	18.7	20.7	23.4	23.8	23.5	23.2	22.4	21.9	19.4	17.6
	2%	DB	22.4	24.6	26.9	29.2	33.3	36.2	38.3	38.9	35.9	31.7	26.3	22.6
		MCWB	14.9	16.4	18.4	20.2	23.0	24.0	23.6	23.4	22.9	21.5	18.5	16.1
	5%	DB	20.1	22.2	25.1	27.6	31.7	35.0	37.1	37.6	34.3	29.7	24.1	20.5
		MCWB	14.6	15.6	17.4	19.6	22.5	23.8	23.8	23.6	22.8	20.3	17.9	15.3
	10%	DB	17.8	20.1	23.1	26.1	30.1	33.7	35.8	36.2	32.9	27.9	22.2	18.1
		MCWB	13.0	14.9	16.6	18.9	22.0	23.7	24.0	23.7	22.4	19.7	17.1	13.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.1	20.0	21.3	23.4	25.7	26.9	26.7	26.2	25.4	24.3	22.0	19.9
		MCDB	21.8	24.0	26.3	28.4	31.6	33.6	33.7	33.5	32.4	28.9	25.8	23.2
	2%	WB	17.8	18.8	20.1	22.2	24.6	25.8	25.8	25.6	24.7	23.3	20.5	18.3
		MCDB	20.3	21.9	24.5	26.9	30.2	32.6	33.4	33.1	31.3	28.2	23.8	21.5
	5%	WB	15.8	17.6	19.1	21.3	23.8	25.2	25.4	25.1	24.1	22.6	19.3	16.5
		MCDB	18.9	20.8	23.2	25.5	29.3	31.8	33.1	32.8	30.4	27.4	22.8	19.2
	10%	WB	13.3	15.7	18.0	20.5	23.1	24.6	25.0	24.7	23.7	21.6	17.9	14.3
		MCDB	16.7	19.4	22.0	24.4	28.4	31.0	32.5	32.3	29.8	26.1	21.6	17.5
Mean Daily Temperature Range	5% DB	MDBR	11.0	11.0	11.0	11.0	10.1	10.1	10.5	10.6	10.5	11.1	10.7	10.4
		MCDBR	14.7	13.9	12.9	12.1	11.5	11.2	11.7	12.0	11.7	12.6	12.4	13.4
		MCWBR	9.0	8.4	6.9	6.3	4.4	3.2	3.0	2.9	3.5	5.2	6.8	8.4
	5% WB	MCDBR	11.6	11.5	10.7	9.9	10.0	10.2	10.3	10.4	9.7	10.0	10.4	11.0
		MCWBR	8.9	8.5	7.2	6.3	5.0	3.7	3.4	3.1	3.5	5.0	7.6	8.8
Clear-Sky Solar Irradiance	taub	0.315	0.328	0.359	0.395	0.427	0.439	0.441	0.442	0.401	0.360	0.338	0.319	
	taud	2.499	2.470	2.402	2.322	2.294	2.294	2.294	2.326	2.307	2.370	2.472	2.473	2.487
	Ebn at Noon	906	927	919	894	864	849	845	841	864	879	869	878	
	Edh at Noon	86	98	113	128	133	133	128	128	116	97	88	83	
All-Sky Solar Radiation	RadAvg	2.88	3.38	4.46	5.41	5.84	6.57	6.70	6.15	5.16	4.05	3.05	2.53	
	RadStd	0.29	0.53	0.34	0.42	0.55	0.63	0.45	0.39	0.45	0.60	0.39	0.32	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.36	N/A	N/A	N/A	-0.45	-0.53	N/A	N/A	+136	+94
(47)	Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page